

Ashwath Karthikeyan

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EDUCATION

University of Illinois at Urbana - Champaign

Master's of Engineering in Robotics

Champaign, IL

Graduation Date: Dec 2024

EXPERIENCE

United Parcel Service

Louisville, KY

Warehouse Robot Data Analyst

Jun 2024 - Present

- Developed a Fleet Telemetry Dashboard to track 200+ warehouse robots simultaneously, identifying operational bottlenecks like deadlocks and de-initialized robots across inbound and outbound pipelines to cut incident response times.
- Wrote an Inventory Reconciliation Service in Python that synchronizes inventory across 3 databases every 5 hours, reporting SKU discrepancies to ensure 99.8% pick accuracy and eliminate robotic dispatch stalls across 50k+ daily items.
- Deployed an Anomaly Detection System to identify orders falling below a specified performance threshold, analyze patterns contributing to delays, and predict the source of issues with 80% accuracy, enhancing overall operational insights.

PROJECTS

Warehouse Fleet Management System

Dec 2025 - Mar 2026

ashwath.net/multirobot/

- Simulated Multi-Robot Path Planning in Python with a live browser viewer, coordinating 12 AGVs across a 60x12 conveyor-fed grid, cutting collision and deadlock rates by 35% via a centralized simultaneous-move conflict resolver.
- Benchmarked pluggable A* and Dijkstra planners across reproducible runs with automated metric logging, improving path optimization by 20% across dynamic warehouse layouts.
- Engineered a Fleet Management System dispatching pickup and drop-off tasks across 10 robots via shared obstacle mapping, achieving 99% avoidance accuracy and cutting resolution times by 20%.

Warehouse Anomaly Scene Generation using LLM

Feb 2025- May 2025

ashwath.net/text2isaacsim/

- Deployed an LLM-driven IsaacSim scene generation interface that constructs custom warehouse environments from text prompts in under 10 seconds, cutting manual simulation setup time by 85%.
- Integrated Ollama with ROS2 to convert input commands into structured scene parameters, achieving 95% accuracy in object placement across 50+ test prompts, improving setup reliability.
- Implemented a ROS communication bridge for dynamically generated environments to coordinate 3+ autonomous robots per scene, achieving a 100% algorithm validation rate across 50 scripted trials.

Automatic Parking in Self-Driving Car

Jan 2024 - May 2024

ashwath.net/self-parking/

- Developed a Full Autonomy Stack in ROS for a self-parking functionality on an autonomous vehicle, enabling real-time detection of empty parking spots and precise navigation into them with a 70% success rate in various parking scenarios.
- Designed and tuned an MPC controller for steering and velocity control, ensuring precise trajectory tracking at speeds up to 20 mph. Designed a Hybrid A* algorithm for efficient real-time route planning with optimized weights.
- Trained a Vision-Based Deep Learning system to identify empty parking spots with 92% mAP. Filtered resulting output to generate a middle-line destination, enabling goal pose estimation for the planning module with an accuracy of 35 cm.

Development of a Rotary Inverted Pendulum

Dec 2023 - Jan 2024

ashwath.net/rip/

- Modeled a Rotary Inverted Pendulum (RIP) control system in MATLAB/Simulink that maintains stable equilibrium across full 360° swing-up trajectories with under 50ms real-time response latency.
- Devised dual control strategies: a basic PID controller effective within 15° of equilibrium, and an advanced state-space controller that expanded the stable operating range to 45°, resulting in a 3x improvement in disturbance rejection capability.
- Formulated a complete mathematical model with 4-dimensional state space, reducing system complexity from nonlinear second-order differential equations to a linear control problem, enabling 95% faster controller tuning and implementation.

SKILLS

Proficiencies: Robot Operating System (ROS/ROS2), Python, C++, MATLAB, OpenCV, SolidWorks

Domains: Warehouse Management Systems (WMS), Warehouse Robotics, Autonomous Vehicles